4 Consider the function $y = \frac{2}{x}$.

a Complete the table of values.

x	-2	-1	$-\frac{1}{2}$	$-\frac{1}{10}$	$-\frac{1}{100}$	$\frac{1}{100}$	$\frac{1}{10}$	$\frac{1}{2}$	1
y									

b For what value of x is the function undefined?

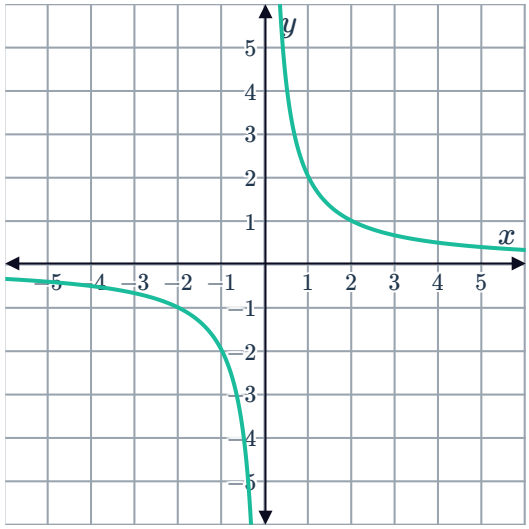
c Rewrite the equation to make x the subject.

d For what value of y is the function undefined?

e Below is the graph of $y = \frac{2}{x}$.
Complete the following statement:

As x approaches ⬚ from the right, the function value approaches ∞. As x approaches ⬚ from the left, the function value approaches $-\infty$.

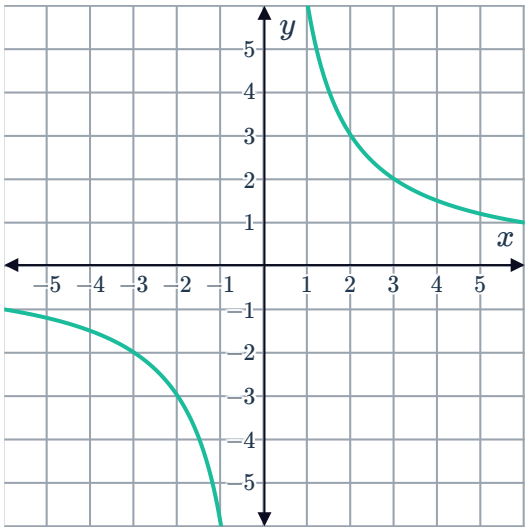
As x approaches ∞ and $-\infty$, y approaches ⬚. This is called the limiting value of the function.



5 Consider the hyperbola shown:

Complete the statement:

"Every point (x, y) on the hyperbola is such that $xy =$ ⬚."



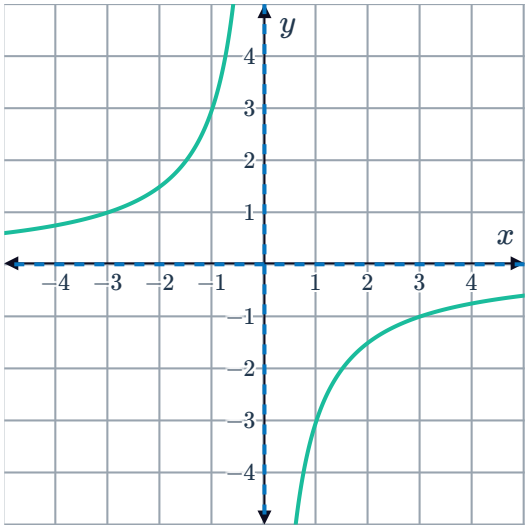
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6 Determine whether the following relationships can be modelled by a function of the form $xy = a$?

- **a** The relationship between the number of people working on a job and how long it will take to complete the job.
- **b** The relationship between the number of sales and the amount of revenue.
- **c** The relationship between height and weight.

7 Consider the function graphed below:

- **a** What is the equation of the vertical asymptote?
- **b** What is the equation of the horizontal asymptote?
- **c** State the domain of the function.
- **d** State the range of the function.



8 For functions of the form $f(x) = \frac{k}{x}$, where k is a constant, the domain is always $(-\infty, 0) \cup (0, \infty)$. What does this mean for the function when $x = 0$?

9 Consider the function $y = \frac{1}{x}$.

- **a** For what value of x is the function undefined?
- **b** What is the domain of $y = \frac{1}{x}$?
- **c** For what value of y is the function underfined?
- **d** What is the range of $y = \frac{1}{x}$?

10 Patricia determined the range of the function $y = \frac{2}{x}$ is $(-\infty, 0) \cup (0, \infty)$.

By filling in the gaps below, complete her reasoning.

Looking at $y = \frac{2}{x}$, notice that the numerator ⬚ is non-zero, and so $\frac{2}{x}$ can never be equal to ⬚.

Another approach is to rearrange the equation and make x the subject. Then we get $x =$ ⬚. In this form we can see that the denominator ⬚ cannot be equal to ⬚.

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11 Consider the functions $y = \frac{4}{x}$ and $y = \frac{2}{x}$.

a When $x = 2$, what is the value of y if $y = \frac{4}{x}$?

b When $x = 2$, what is the value of y if $y = \frac{2}{x}$?

c Which graph lies further away from the axes?

d For hyperbolas of the form $y = \frac{k}{x}$, as the k value increases, what happens to the graph?

Graphs of hyperbolas

12 Consider the function $y = \frac{6}{x}$.

a Complete the following tables of values:

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y					

b Plot the points from the table of values on a cartesian plane.

13 Ursula wants to sketch the graph of $y = \frac{7}{x}$, but knows that it will look similar to many other hyperbolas.

What can she do to the graph to show that it is the hyperbola $y = \frac{7}{x}$, rather than any other hyperbola of the form $y = \frac{k}{x}$?

14 For each of the following functions:

i Complete a table of values of the form:

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y						

ii Sketch the graph of the function.

iii State which quadrants the function lies in.

a $y = \frac{2}{x}$ **b** $y = -\frac{1}{x}$ **c** $y = -\frac{1}{4x}$

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Applications

15 The relationship between the current, C, (in amperes) and resistance, R, (in ohms) in an electrical circuit is given by:

$$C = \frac{240}{R}$$

where the voltage provided to the circuit is 240 V.

a Graph this function for $0 \leq R \leq 50$.

b What happens to the current as the resistance increases?

16 The rent, electricity, telephone bill and other expenses for a flat costs a total of $\$490$ per week. These expenses are shared equally between the tenants of the flat.

a How much will each occupant pay in dollars if the flat is shared by two people?

b Let the number of occupants be x, and the cost paid by each occupant be y.

Write a formula that relates the two pronumerals.

c Use this equation to fill in the following table of values.

x	1	2	3	4	5	6
y						

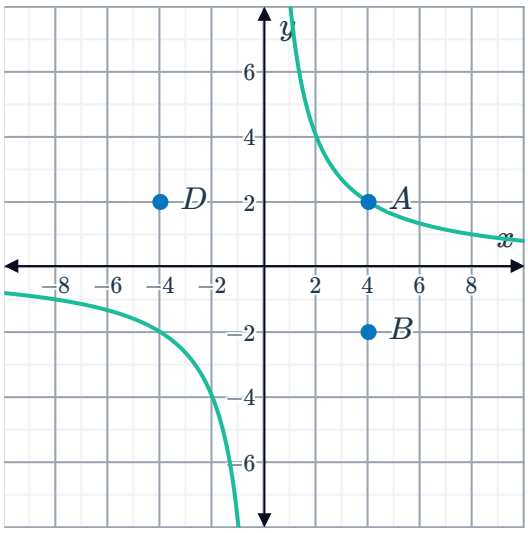
d Sketch the curve from your equation.

e What type of relationship exists between x and y?

17 Consider the hyperbola that has been graphed. Points $A\,(4, 2)$, B, $C\,(-4, k)$ and D form the vertices of a rectangle.

a Find the value of k.

b Hence find the area of the rectangle with vertices $ABCD$.



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18 Boyle's law describes the relationship between pressure and volume of a gas of fixed mass under constant temperature. The pressure for a particular gas can be found using

$$P = \frac{6000}{V}$$

where P has units kg/cm^2 and V has units cm^3.

a Graph the relationship $P = \dfrac{6000}{V}$ for $0 \leq V \leq 2000$.

b What is the pressure if the volume is $1\ \text{cm}^3$?

c What happens to the pressure as the volume increases?

19 The time it takes a commuter to travel $100\ \text{km}$ depends on how fast they are going. We can write this using the equation $t = \dfrac{100}{S}$ where S is the speed in km/h and t is the time taken in hours.

a Graph the relationship $t = \dfrac{100}{S}$.

b What is the time taken if the speed travelled is $10\ \text{km/h}$?

c What is the time taken if the speed travelled is $50\ \text{km/h}$?

d If we want the travel time to decrease, what must happen to the speed of travel?

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